

Assignment 2

Course : Cyber Security (TCS 495)

Units Covered : Unit 3 and Unit 4

Total Marks : 50

Submission Mode : Handwritten Report + Practical
Screenshots + Program Files

Submission Date : 6 PM 30/04/2026 Faculty Cabin

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Q1. Explain the importance of computer networking in Ethical Hacking.  
Why must a cybersecurity student understand TCP/IP? Give five reasons.

Ans:

Computer networking is the foundation of all communication in a network. Ethical hackers must know how data travels, how devices communicate and which protocols are used. Without networking knowledge, it is impossible to scan, sniff, exploit or secure a network.

Five reasons to understand TCP/IP :

- 1) It is the base protocol suite of the internet.
- 2) Helps in understanding packet flow and communication.
- 3) Required for network scanning, enumeration and traffic analysis.
- 4) Helps in troubleshooting and identifying vulnerabilities.
- 5) Essential for tools like Wireshark, Nmap, Metasploit etc.

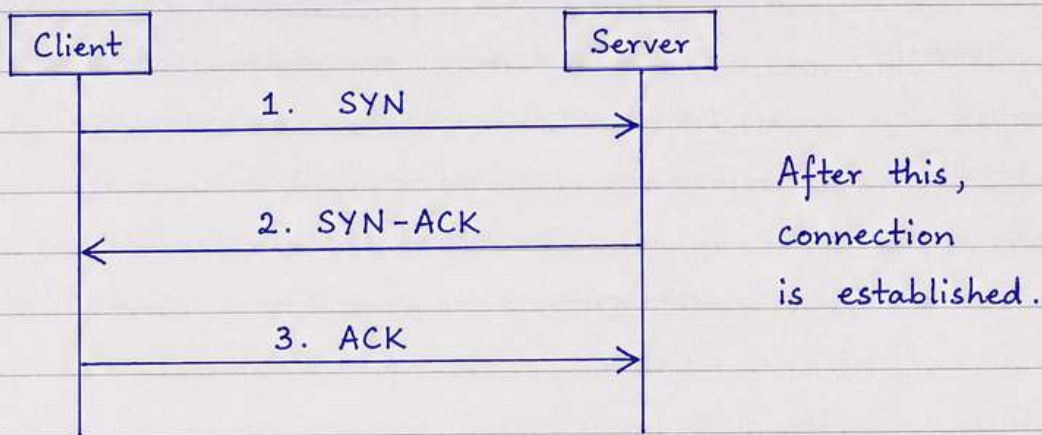
Q2. Differentiate between the following virtual machine network modes in tabular form and mention one real use case of each.

Ans:

| Mode      | How it Works                       | Host Access | External Access | Use Case                                         |
|-----------|------------------------------------|-------------|-----------------|--------------------------------------------------|
| NAT       | VM shares host IP using NAT        | Yes         | Yes (via host)  | VMs access internet but not visible from outside |
| Bridged   | VM acts like another device on LAN | Yes         | Yes (direct)    | Testing web server or when VM must be visible    |
| Host-Only | VMs communicate with host only     | Yes         | No              | Isolated testing & lab environment               |

Q3. Explain the TCP 3-Way Handshake with diagram and packet flow. What risks arise if handshake security is weak?

Ans:



- 1) Client sends SYN (Synchronize) to server.
- 2) Server replies with SYN-ACK.
- 3) Client sends ACK. Connection established.

Risks if handshake security is weak :

- Attackers can perform SYN flood attacks.
- Spoofed connections can be established.
- Leads to Denial of Service (DoS).
- Session hijacking can occur.

Q4. Explain the difference between :

Ans: 1) IPv4 vs IPv6

|               | IPv4                  | IPv6               |
|---------------|-----------------------|--------------------|
| Address size  | 32-bit address        | 128-bit address    |
| Notation      | Dotted decimal        | Hexadecimal        |
| Example       | 192.168.1.1           | 2001:db8::1        |
| NAT           | NAT required          | NAT not required   |
| Address space | Limited address space | Huge address space |

2) IP Address vs MAC Address

- IP Address : Logical address, changes, works at network layer.
- MAC Address : Physical address, unique to device, works at data link layer.

3) DNS vs Ping

- DNS : Resolves domain name to IP address.
- Ping : Checks reachability of a host.

Q5. A company is assigned IP address 192.168.10.0/24 and needs 4 departments with equal hosts. Find :

- a) New subnet mask
- b) Number of subnets created
- c) Number of usable hosts per subnet
- d) First 4 subnet addresses

Ans: Given : 192.168.10.0/24      Need 4 equal subnets

- a) 2 bits needed to borrow because  $2^2 = 4$

$$\text{New prefix} = /24 + 2 = /26$$

$$\text{New subnet mask} = 255.255.255.192$$

- b) Number of subnets created =  $2^2 = 4$

- c) Host bits remaining =  $32 - 26 = 6$

$$\text{Usable hosts per subnet} = 2^6 - 2 = 64 - 2 = 62$$

- d) First 4 subnet addresses :

1) 192.168.10.0 /26

2) 192.168.10.64 /26

3) 192.168.10.128 /26

4) 192.168.10.192 /26

Q6. Given IP address 172.16.25.130/26, determine :

- a) Network address
- b) Broadcast address
- c) Usable host range
- d) Total usable hosts

Ans: /26  $\Rightarrow$  block size = 64

Subnets in last octet : 0, 64, 128, 192

IP 130 falls in 128 - 191 range

a) Network address : 172.16.25.128

b) Broadcast address : 172.16.25.191

c) Usable host range : 172.16.25.129 - 172.16.25.190

d) Total usable hosts :  $2^{(32-26)} - 2 = 62$

Q7. Install any virtualization software (VirtualBox) and create one Linux VM. Write steps with screenshots.

Ans:

- 1) Installation : Download VirtualBox from official website → run installer → follow setup wizard → finish.
- 2) VM creation : Click "New" → Give name (Kali Linux) → Type : Linux → Version : Debian (64-bit) → Next.
- 3) RAM and storage allocation :
  - Base Memory : 2048 MB (or as per requirement)
  - Create a virtual hard disk → VDI → Dynamically allocated → Size : 20 GB → Create.
- 4) Change network adapter to NAT :
  - Go to Settings → Network → Adapter 1 → Attached to : NAT → OK.
- 5) Change network adapter to Bridged :
  - Go to Settings → Network → Adapter 1 → Attached to : Bridged Adapter → select your physical network adapter → OK.



← Create Virtual Machine

### Name and operating system

Please choose a descriptive name and destination folder for the new virtual machine and select the type of operating system you intend to install on it. The name you choose will be used throughout VirtualBox to identify this machine.

Name:

Kali Linux

Folder:

C:\VirtualBox VMs

Type:

Linux

Version:

Debian (64-bit)



Expert Mode

Next

Cancel



the left part of application  
g tools and lists all virtual  
n. You can import, add and  
, corresponding toolbar  
button.  
press the **F1** key to get  
information.

Tools

New

Settings

Discard

Start

?

×

←

Create Virtual Machine

Memory size

Select the amount of memory (RAM) in megabytes to be allocated to the virtual machine.  
The recommended memory size is **1024 MB**.

4 MB16384 MB

2048

MB

Hard disk

☐ Do not add a virtual hard disk

☒ Create a virtual hard disk now

☐ Use an existing virtual hard disk file

Empty

Help

Back

Next

Cancel



global tools and  
create, add and  
corresponding toolbar

ou can press the **F1**  
formation and latest  
s.

←

Create Virtual Hard Disk

?

×

Hard disk file type

Please choose the type of file that you would like to use for the new virtual hard disk. If you do not need to use it with other virtualization software you can leave this setting unchanged.

☒

VDI (VirtualBox Disk Image)

☐

VHD (Virtual Hard Disk)

☐

VMDK (Virtual Machine Disk)

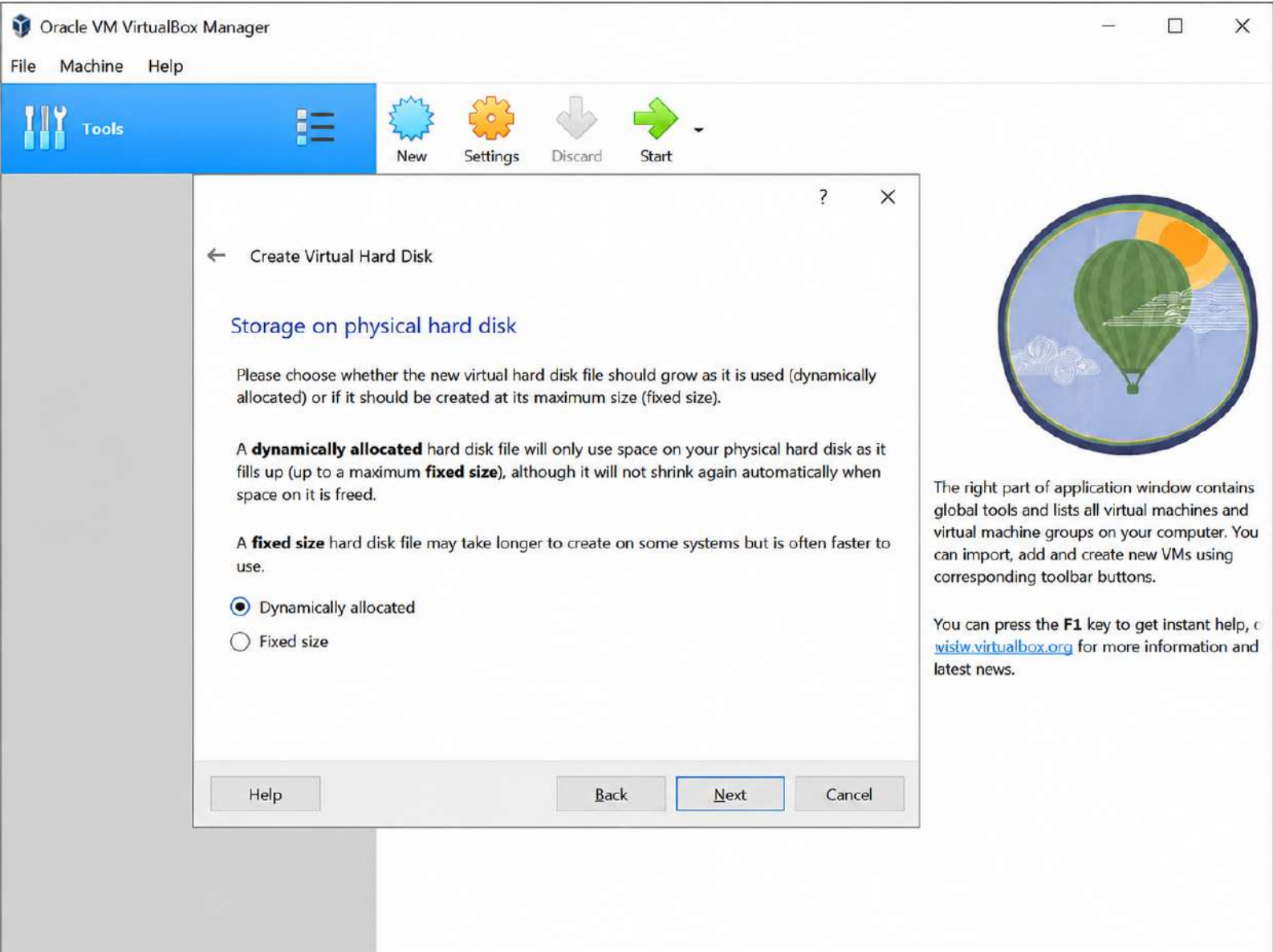
Expert Mode

Next

Cancel



the left part of application  
p tools and lists all virtual  
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, corresponding toolbar  
button.  
press the **F1** key to get  
information.



Kali Linux - Settings

General System Display Storage Audio **Network** Serial Ports USB Shared Folders User Interface

### Network

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to: NAT

Name:

Advanced

Adapter Type: Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode: Deny

MAC Address: 08002734D7E4

☒ Cable Connected

Port Forwarding

OK Cancel Help

Kali Linux - Settings

General System Display Storage Audio **Network** Serial Ports USB Shared Folders User Interface

### Network

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to: Bridged Adapter

Name: Realtek PCIe GbE Family Controller

Advanced

Adapter Type: Intel PRO/1000 MT Desktop (82540EM)

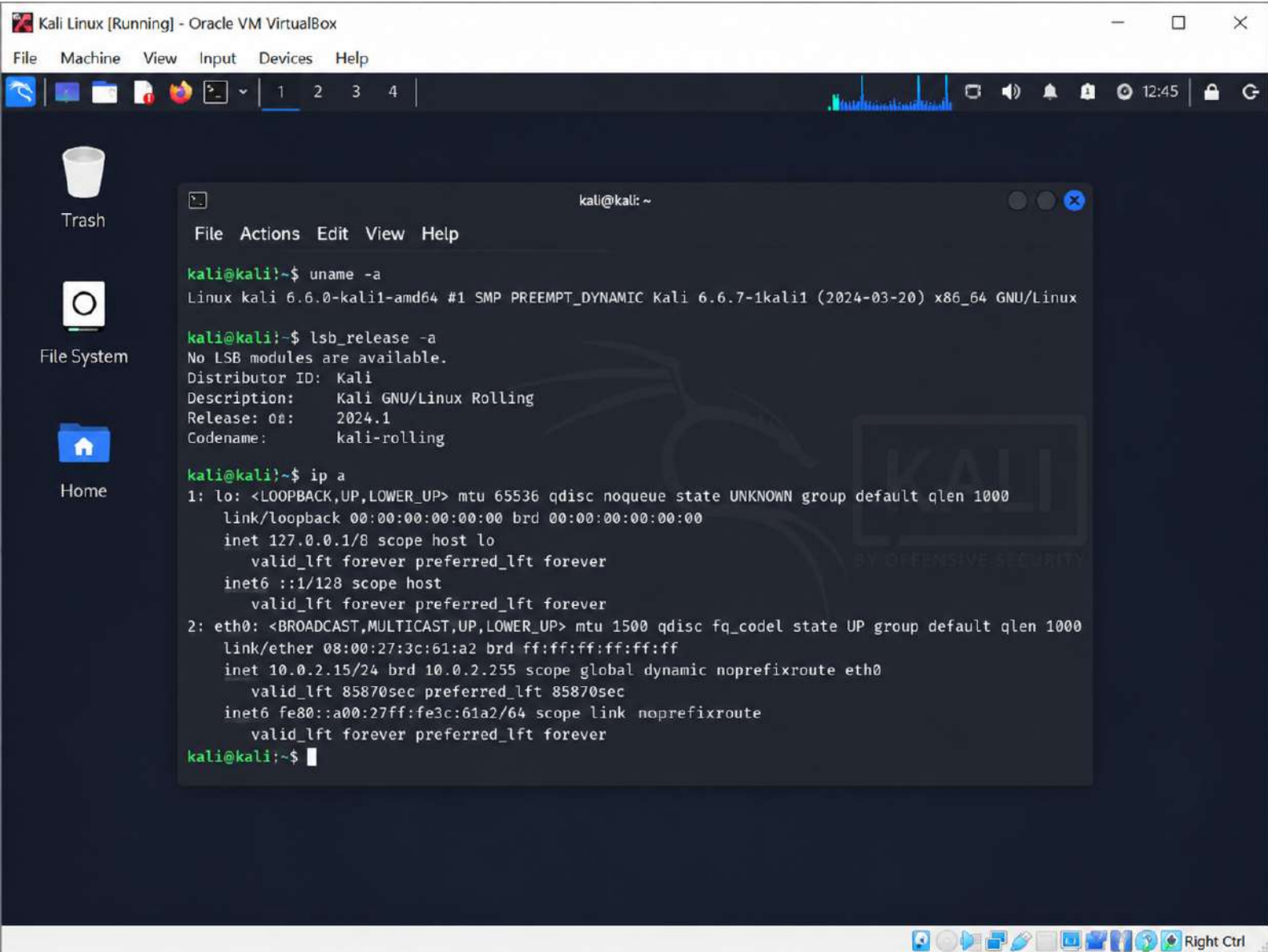
Promiscuous Mode: Deny

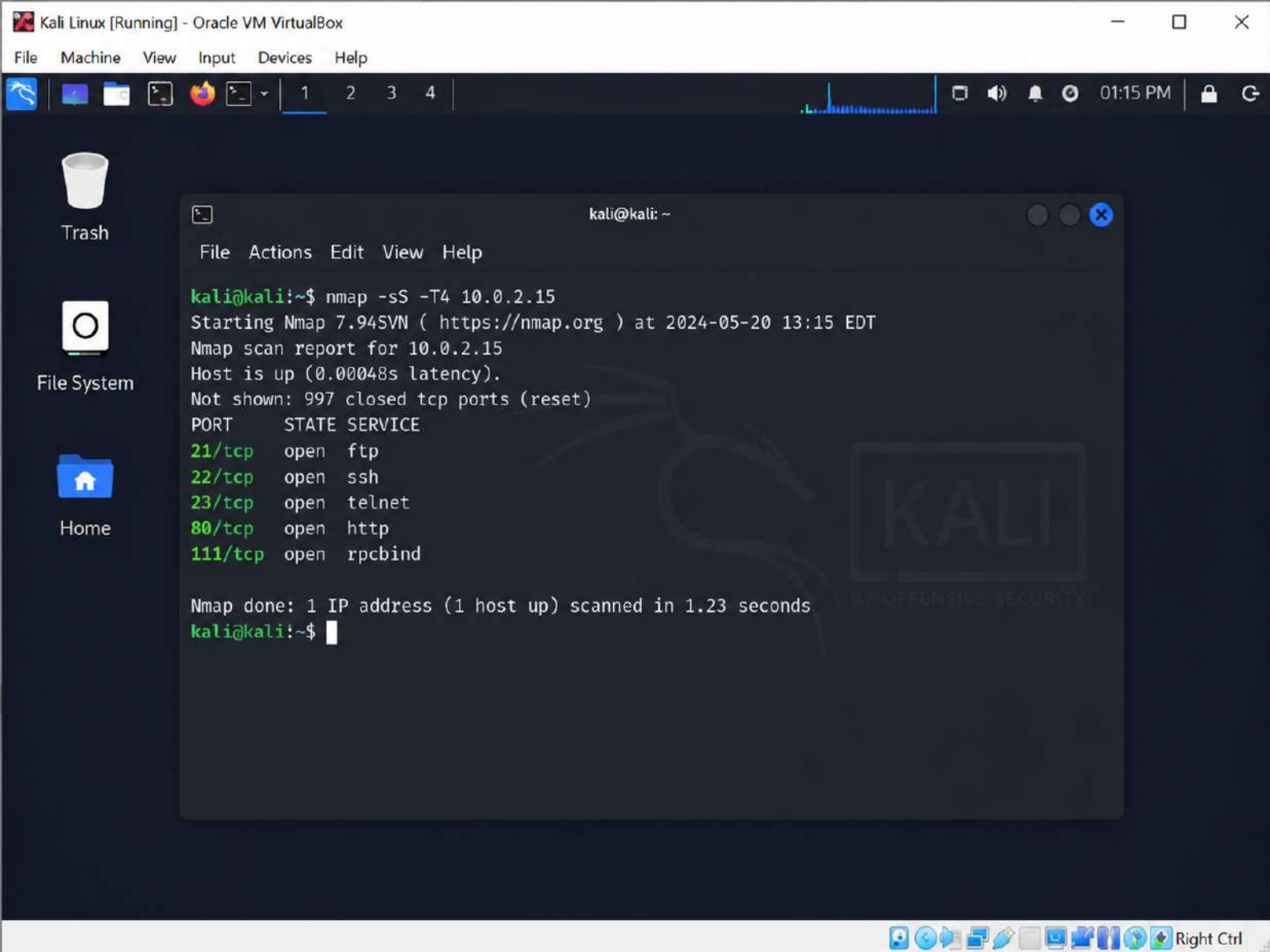
MAC Address: 08002737C61A

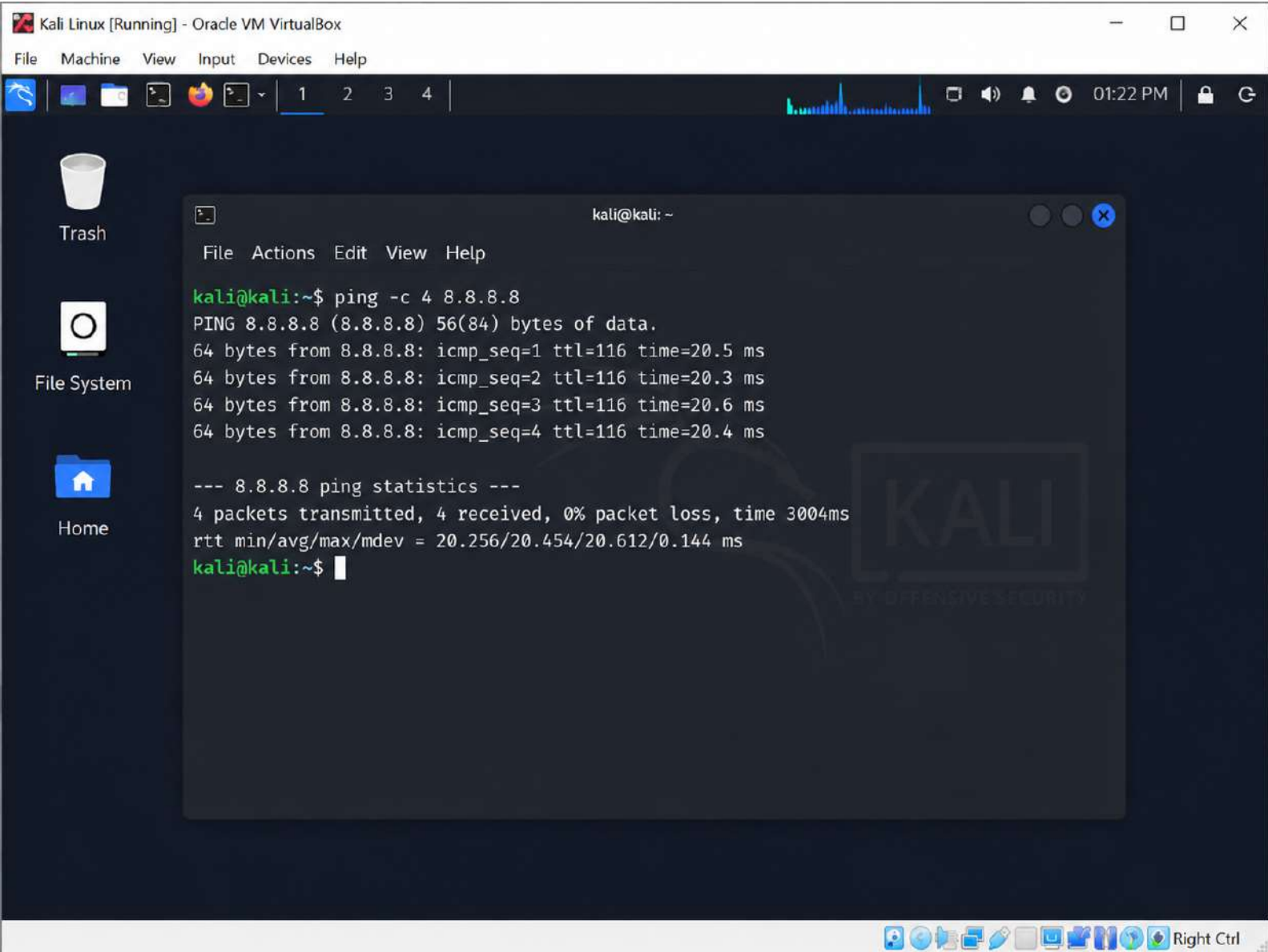
☒ Cable Connected

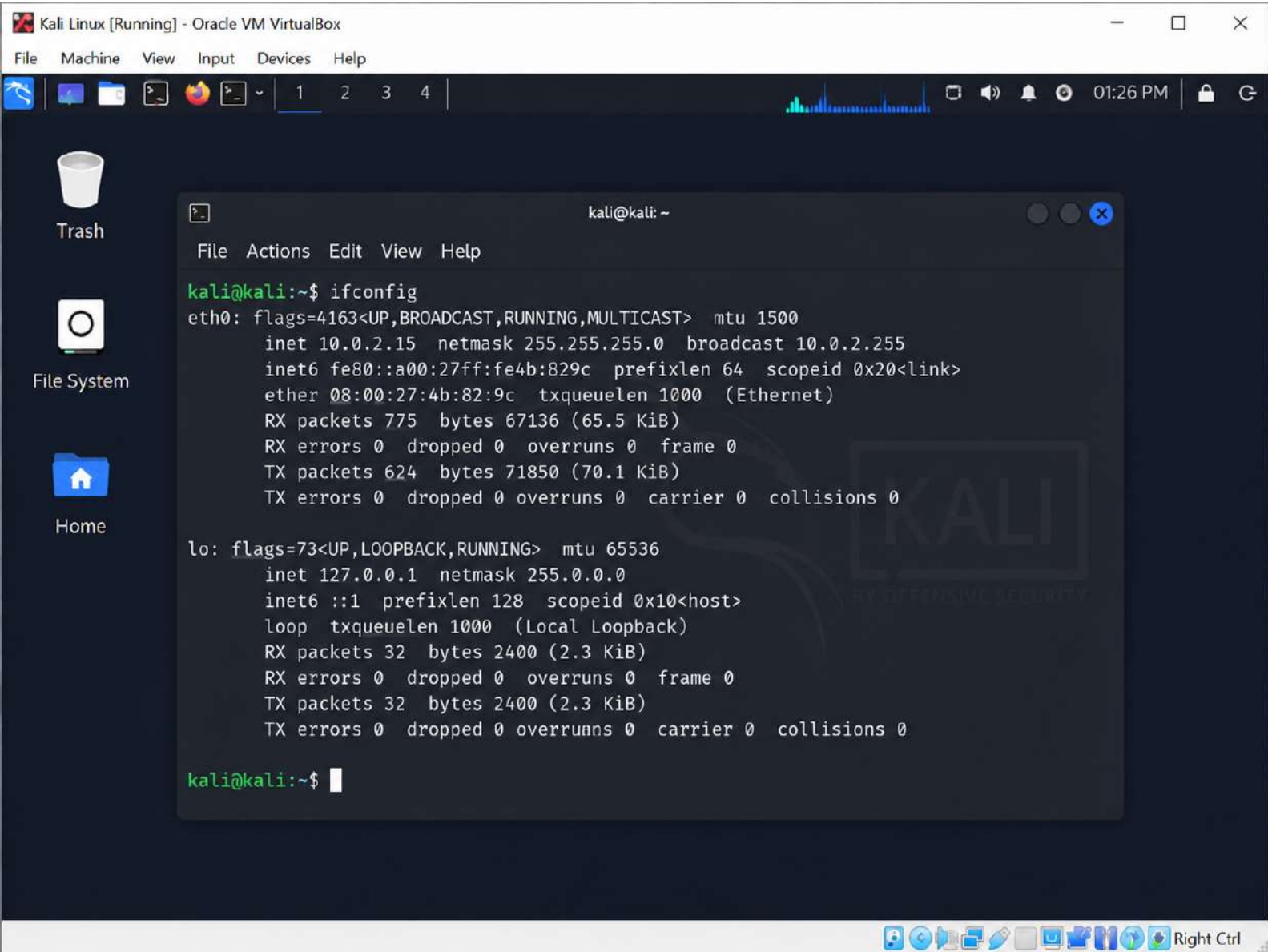
Port Forwarding

OK Cancel Help









Q8. Explain Wireshark and perform packet capture on your machine. Capture DNS or Ping traffic and answer:

- a) Source IP
- b) Destination IP
- c) Protocol used
- d) Packet count

Attach screenshot.

Ans: Wireshark is a network protocol analyzer that captures and displays packets travelling over a network. It helps in troubleshooting, security analysis and understanding network traffic.

Captured Ping Traffic :

| No. | Source IP   | Destination IP | Protocol | Info                |
|-----|-------------|----------------|----------|---------------------|
| 1   | 192.168.1.6 | 8.8.8.8        | ICMP     | Echo (ping) request |
| 2   | 8.8.8.8     | 192.168.1.6    | ICMP     | Echo (ping) reply   |
| 3   | 192.168.1.6 | 8.8.8.8        | ICMP     | Echo (ping) request |
| 4   | 8.8.8.8     | 192.168.1.6    | ICMP     | Echo (ping) reply   |
| 5   | 192.168.1.6 | 8.8.8.8        | ICMP     | Echo (ping) request |

- a) Source IP : 192.168.1.6
- b) Destination IP : 8.8.8.8
- c) Protocol used : ICMP
- d) Packet count : 5

Wireshark

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl+>

| No. | Time     | Source      | Destination | Protocol | Length | Info                                              |
|-----|----------|-------------|-------------|----------|--------|---------------------------------------------------|
| 1   | 0.000000 | 192.168.1.6 | 8.8.8.8     | ICMP     | 98     | Echo (ping) request id=0x0001, seq=1/256, ttl=64  |
| 2   | 0.021123 | 8.8.8.8     | 192.168.1.6 | ICMP     | 98     | Echo (ping) reply id=0x0001, seq=1/256, ttl=117   |
| 3   | 1.000415 | 192.168.1.6 | 8.8.8.8     | ICMP     | 98     | Echo (ping) request id=0x0001, seq=2/512, ttl=64  |
| 4   | 1.021987 | 8.8.8.8     | 192.168.1.6 | ICMP     | 98     | Echo (ping) reply id=0x0001, seq=2/512, ttl=117   |
| 5   | 2.000623 | 192.168.1.6 | 8.8.8.8     | ICMP     | 98     | Echo (ping) request id=0x0001, seq=3/768, ttl=64  |
| 6   | 2.022194 | 8.8.8.8     | 192.168.1.6 | ICMP     | 98     | Echo (ping) reply id=0x0001, seq=3/768, ttl=117   |
| 7   | 3.000834 | 192.168.1.6 | 8.8.8.8     | ICMP     | 98     | Echo (ping) request id=0x0001, seq=4/1024, ttl=64 |
| 8   | 3.022352 | 8.8.8.8     | 192.168.1.6 | ICMP     | 98     | Echo (ping) reply id=0x0001, seq=4/1024, ttl=117  |

> Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF\_{8C2B6E1E-2F1B-4E7A-A3A1-3F0D6C5B2D3E}, id 0

> Ethernet II, Src: IntelCor\_3a:5d:7b (88:36:6c:3a:5d:7b), Dst: Tp-LinkTe\_2b:1a:c3 (f0:9f:c2:2b:1a:c3)

> Internet Protocol Version 4, Src: 192.168.1.6, Dst: 8.8.8.8

> Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0xf7ff [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

> Data (56 bytes)

```
0000 f0 9f c2 2b 1a c3 88 36 6c 3a 5d 7b 08 00 45 00 ...+....61 :]{...E.
0010 00 54 1c 46 40 00 40 01 5c d7 c0 a8 01 06 08 08 .T.F@.@.\. ....
0020 08 08 00 08 f7 ff 00 01 00 01 61 62 63 64 65 66 ..... !,abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmnopqrstuv
0040 77 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f wabcdeffghi.....
0050 70 71 72 73 74 75 76 77 00 01 02 03 pqrstuvwxyz .....
```

wireshark\_1C8I24LCA3.pcapng

Packets: 8 · Displayed: 8 (100.0%)

Profile: Default

Q9. Demonstrate basic client-server communication using Netcat (nc).

Ans:

Netcat (nc) is a powerful networking utility that can be used to read from and write to network connections using TCP or UDP. It can act as both a client and a server. Below is a demonstration of basic client-server communication using Netcat over TCP.

Step 1: Open two terminal windows in Kali Linux.

Step 2: On the server terminal, start Netcat in listening mode on port 4444.

```
nc -l -p 4444 → (Server)
```

Step 3: On the client terminal, connect to the server IP (127.0.0.1) on port 4444.

```
nc 127.0.0.1 4444 → (Client)
```

Step 4: Once the connection is established, text typed in the client terminal is received by the server and vice-versa. This demonstrates basic client-server communication.

Note: Here, both server and client are running on the same Kali Linux machine using localhost (127.0.0.1).

Conclusion:

The above demonstration shows successful client-server communication using Netcat (nc) over TCP on port 4444.

```
kali@kali: ~  
File Actions Edit View Help  
kali@kali:~$ nc -l -p 4444  
listening on [any] 4444 ...  
connect to [127.0.0.1] from (UNKNOWN) [127.0.0.1] 49784  
Hello Server, this is Client!  
How are you?  
I am fine. This is Netcat demo.  
Great! Netcat is working perfectly.  
Yes, it works!  
Bye from Client.  
Bye! Connection closed.  
  
kali@kali:~$
```

```
kali@kali: ~  
File Actions Edit View Help  
kali@kali:~$ nc 127.0.0.1 4444  
Hello Server, this is Client!  
How are you?  
I am fine. This is Netcat demo.  
Great! Netcat is working perfectly.  
Yes, it works!  
Bye from Client.  
  
kali@kali:~$
```

Q10.

Write a PHP program that connects to SQL database and performs secure login validation.

```
<?php
// Database connection using MySQLi with prepared statements
$conn = new mysqli("localhost", "root", "", "login_db");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
// Check if form is submitted
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $username = $_POST['username'];
    $password = $_POST['password'];
    // Prepare and bind
    $stmt = $conn->prepare("SELECT id, username FROM users
                            WHERE username = ? AND password = ?");
    $stmt->bind_param("ss", $username, $password);
    $stmt->execute();
    $result = $stmt->get_result();
    // Validate login
    if ($result->num_rows == 1) {
        echo "Login successful! Welcome, " . $username;
    } else {
        echo "Invalid username or password!";
    }
    $stmt->close();
    $conn->close();
}
?>
```

a) What is SQL Injection?

SQL Injection is an attack technique where an attacker inserts malicious SQL code into input fields to manipulate a database query and access, modify or delete data without proper authorization.

b) How prepared statements prevent it?

Prepared statements separate SQL code from user input. The user input is treated strictly as data, not as executable code. This prevents attackers from injecting malicious SQL commands into the query.

c) Difference between phishing and spear phishing

| Phishing                                                                                                                                                                                                                     | Spear Phishing                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• General attack sent to many people.</li><li>• Uses generic messages.</li><li>• Targets a wide audience.</li><li>• Example: Fake emails to many users asking for passwords.</li></ul> | <ul style="list-style-type: none"><li>• Targeted attack against a specific person or organization.</li><li>• Uses personalized information.</li><li>• Targets a specific individual.</li><li>• Example: Email to an employee with details about their company.</li></ul> |

Bonus Challenge

Create a mini project titled Cyber Network Detective using any language/tool that performs any three tasks:

- Ping scanner
- Port checker
- IP calculator
- DNS lookup tool
- Form validator